

ROTATIONAL DYNAMICS

Formula • Derivation • Conditions • Master Atlas

Complete derivation companion for JEE Main & JEE Advanced (Mechanics)

Sign convention used throughout: counter-clockwise (CCW) sense of a vector quantity (angular displacement θ , angular velocity $\vec{\omega}$, angular acceleration $\vec{\alpha}$, torque $\vec{\tau}$, angular momentum $\vec{L}$) is taken *positive* and is directed out of the plane of rotation by the right-hand rule; clockwise (CW) is negative. Newton's second law for rotation: $\tau_{net} = I\alpha$. All boxed results are final, exam-ready formulas; every box is preceded by its derivation and followed by the conditions under which it applies.

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Part I — Foundations of Rigid Body Rotation

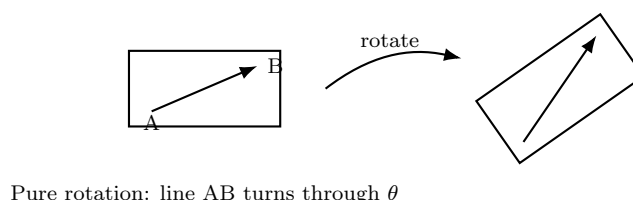
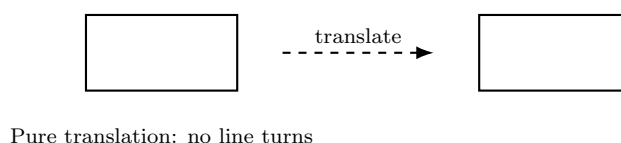
1 Rigid Body and Types of Motion

1.1 Rigid Body

A **rigid body** is a system of particles in which the distance between every pair of particles remains constant under the action of any force. Real bodies deform, but when the deformation is negligible for the motion under study, the body is idealised as rigid.

Any motion of a rigid body is a combination of:

- **Pure Translation** — every particle has the same displacement, velocity and acceleration at a given instant; orientation of the body does not change.
- **Pure Rotation** — particles move on circles about a fixed line (the axis); orientation changes.
- **General (Combined) Plane Motion** — superposition of translation of a reference point (usually the centre of mass) and rotation about that point.



1.2 Axis of Rotation

An imaginary line, perpendicular to the plane of the circular paths of all particles and passing through the centres of these circular paths, is the **axis of rotation**. It need not pass through the body itself (e.g. a block glued off-centre on a rotating disc still revolves about the disc's central axis).

Classification of rotational motion:

1. **Rotation about a fixed axis** — ceiling fan, door on hinges; the axis itself is stationary.
2. **Rotation about an axis in translation** — rolling wheel, a rod sliding and turning on a table (general plane motion); the axis translates while the body spins about it.
3. **Rotation about an axis in rotation** — precession of a spinning top (outside JEE scope, mentioned for completeness).

When & How to Use

Identify which of the three categories a problem belongs to *first*. Fixed-axis problems need only $\tau = I\alpha$ about that axis. Axis-in-translation (rolling/general plane motion) problems need *two* equations: Newton's second law for the centre of mass, $F_{net} = Ma_{cm}$, and the torque equation about the centre of mass, $\tau_{cm} = I_{cm}\alpha$ (or a single equation about the ICR/point of contact when permissible, see Part IV).

2 Angular Displacement, Velocity and Acceleration

2.1 Definitions

Let θ be the angle turned by a reference line on the body.

Formula 1.1

Angular velocity and angular acceleration

$$\vec{\omega} = \frac{d\vec{\theta}}{dt}, \quad \vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{d^2\vec{\theta}}{dt^2} = \omega \frac{d\omega}{d\theta}$$

Conditions θ is measured in radians. $\vec{\omega}$ and infinitesimal $d\vec{\theta}$ are vectors along the axis of rotation, direction given by the right-hand (curl) rule. $\vec{\alpha}$ points along $\vec{\omega}$ if speed is increasing, and opposite to $\vec{\omega}$ if speed is decreasing. For a fixed (non-rotating) axis, ω, α may be treated as signed scalars.

2.2 Equations of Rotational Kinematics (constant α)

Derivation. Exactly analogous to $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$ of rectilinear motion, obtained by direct integration of $\alpha = d\omega/dt = \text{constant}$ and $\omega = d\theta/dt$.

Formula 1.2

Rotational kinematic equations

$$\omega = \omega_0 + \alpha t \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2 = \left(\frac{\omega_0 + \omega}{2}\right)t \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

Conditions Valid only when angular acceleration α is constant, exactly parallel to the equations of uniformly accelerated rectilinear motion with $\theta \leftrightarrow s$, $\omega \leftrightarrow v$, $\alpha \leftrightarrow a$.

2.3 Relation between Linear and Angular Quantities (fixed axis)

For a particle at perpendicular distance r from a fixed axis:

Formula 1.3

Linear-angular relations for a particle on a rotating rigid body

$$v = \omega r \quad a_t = \alpha r \text{ (tangential)} \quad a_n = \omega^2 r = \frac{v^2}{r} \text{ (centripetal/normal)}$$

$$\vec{v} = \vec{\omega} \times \vec{r} \quad \vec{a}_t = \vec{\alpha} \times \vec{r} \quad \vec{a}_n = \vec{\omega} \times \vec{v} = -\omega^2 \vec{r}$$

Conditions r is the perpendicular distance from the fixed axis, not the distance from the origin. a_t changes the speed; a_n changes the direction only. Net acceleration $\vec{a} = \vec{a}_t + \vec{a}_n$, with $|\vec{a}| = r\sqrt{\alpha^2 + \omega^4}$.

Common Trap Radial (centripetal) force never contributes torque about the axis it points towards (it passes through/intersects the axis); only the tangential force $F_t = ma_t = mra$ contributes torque $= F_t r = mr^2\alpha$. Summing over all particles gives $\tau = I\alpha$ (Part III).

Part II — Moment of Inertia

3 Moment of Inertia: Definition and Dependence

Moment of inertia (M.I.) is the rotational analogue of mass: it measures a body's opposition to a *change* in its rotational motion produced by a torque, exactly as mass measures opposition to a change in translational motion produced by a force.

Moment of inertia of a body about a given axis depends on:

1. Mass of the body (and the material density distribution).
2. Shape and size of the body.
3. Position and orientation of the axis of rotation.

Combined, M.I. depends on how mass is *distributed relative to the axis*. Unlike mass, M.I. is not an intrinsic property of a body – it is always quoted *about a stated axis*.

3.1 Moment of Inertia of a System of Particles

Formula 2.1

M.I. of a single particle and of a system of particles

$$I_{\text{particle}} = mr^2 \qquad I_{\text{system}} = \sum_{i=1}^n m_i r_i^2$$

Conditions r_i is the *perpendicular* distance of the i^{th} particle from the axis (not from a point). SI unit: kg m^2 ; dimension $[ML^2]$.

3.2 Moment of Inertia of a Continuous (Rigid) Body

Derivation. A rigid body is treated as a continuum of infinitesimal mass elements dm . Each element at perpendicular distance r from the axis contributes $dI = r^2 dm$. Summing (integrating) over the entire body:

Formula 2.2

General integral definition of M.I.

$$I = \int dI = \int r^2 dm$$

Conditions Choose dm using the relevant mass density: linear $\lambda = dm/dx$ for rods/rings/arcs, surface $\sigma = dm/dA$ for plates/discs, volume $\rho = dm/dV$ for solids. For a *uniform* body λ, σ, ρ are constants equal to (total mass)/(total length, area or volume); for a *non-uniform* body they are functions of position and must be kept inside the integral (Section 5.3).

3.3 Radius of Gyration (k)

Derivation. Radius of gyration is defined as the distance from the axis at which, if the *entire* mass of the body were concentrated as a point mass, the M.I. would be unchanged: $I = Mk^2$.

Formula 2.3

Radius of Gyration

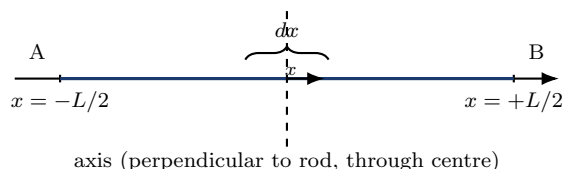
$$k = \sqrt{\frac{I}{M}} \qquad \Longleftrightarrow \qquad I = Mk^2$$

Conditions k has dimensions of length. It depends on the axis chosen (like I) and on how mass is distributed, but *not* on the actual value of the mass M itself. Useful shortcut form for rolling-body problems: write every standard M.I. as $I_{cm} = Mk^2$ (e.g. $k^2 = R^2/2$ for a disc, $k^2 = 2R^2/5$ for a solid sphere).

Key Points Moment of inertia is a scalar for rotation about a fixed axis; in general (3-D, tumbling) motion it is represented by a tensor, which is outside the JEE syllabus. M.I. does *not* change if mass is rearranged along a circle of the same radius about the same axis.

4 Moment of Inertia of a Rod

4.1 Straight Uniform Rod — about the Centre (perpendicular to rod)



Derivation. Let the uniform rod have mass M , length L , so linear mass density $\lambda = M/L$. Take the origin at the centre, with the axis perpendicular to the rod through this point. Consider an element of length dx at distance x from the axis; its mass is $dm = \lambda dx$.

$$I = \int_{-L/2}^{L/2} x^2 \lambda dx = \lambda \left[\frac{x^3}{3} \right]_{-L/2}^{L/2} = \lambda \cdot \frac{2}{3} \left(\frac{L}{2} \right)^3 = \frac{\lambda L^3}{12} = \frac{M}{L} \cdot \frac{L^3}{12}$$

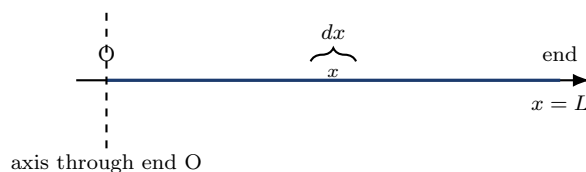
Formula 2.4

M.I. of a uniform rod about the centroidal axis $\perp$ to rod

$$I_C = \frac{ML^2}{12}$$

Conditions Rod is thin (cross-section negligible), mass uniformly distributed ($\lambda = \text{const}$), axis passes through the mid-point and is perpendicular to the length. This is the *minimum* M.I. among all axes parallel to it (see Parallel Axis Theorem, Sec. 8).

4.2 Uniform Rod — about One End (perpendicular to rod)



Derivation. Take the origin at one end. An element $dm = \lambda dx = \frac{M}{L} dx$ lies at distance x from the axis, $0 \leq x \leq L$.

$$I = \int_0^L x^2 \frac{M}{L} dx = \frac{M}{L} \left[\frac{x^3}{3} \right]_0^L = \frac{ML^2}{3}$$

Formula 2.5

M.I. of a uniform rod about an axis through one end, $\perp$ to rod

$$I_{\text{end}} = \frac{ML^2}{3}$$

Conditions Same physical rod as Sec. 4.1; can also be obtained directly from Eq. 2.4 using the Parallel Axis Theorem: $I_{end} = I_C + M(L/2)^2 = \frac{ML^2}{12} + \frac{ML^2}{4} = \frac{ML^2}{3}$ – a useful cross-check.

4.3 Non-Uniform Rod (variable linear density)

Derivation. When mass per unit length is *not* constant, $\lambda = \lambda(x)$ must be kept inside the integral, and total mass is obtained first by integrating $\lambda(x)$ itself. As a representative case take $\lambda(x) = kx$ (density increasing linearly from one end, a frequently asked JEE variant), rod of length L lying from $x = 0$ to $x = L$, axis through the end $x = 0$.

$$M = \int_0^L \lambda(x) dx = \int_0^L kx dx = \frac{kL^2}{2} \Rightarrow k = \frac{2M}{L^2}$$

$$I_{end} = \int_0^L x^2 \lambda(x) dx = \int_0^L kx^3 dx = \frac{kL^4}{4} = \frac{2M}{L^2} \cdot \frac{L^4}{4}$$

Formula 2.6

M.I. of a non-uniform rod, $\lambda(x) = kx$, about the end $x = 0$

$$I_{end} = \frac{ML^2}{2} \quad \left(\text{general recipe: } M = \int \lambda(x) dx, \quad I = \int x^2 \lambda(x) dx \right)$$

Conditions The exact numerical coefficient depends entirely on the given $\lambda(x)$ – always re-derive, never memorise a single non-uniform answer. The two-step method (find M first in terms of k , then find I , then eliminate k) is the general procedure for *any* $\lambda(x)$: linear, quadratic, or defined piecewise.

Common Trap For a non-uniform body, the geometric centre and the centre of mass do *not* coincide, so the Parallel Axis Theorem must use the actual distance to the *centre of mass*, which itself must be computed by $x_{cm} = \frac{1}{M} \int x dm$ – it is *not* automatically $L/2$.

4.4 Straight Uniform Rod — General Axis (summary)

For a straight (unbent) uniform rod of mass M , length L , the M.I. about any axis perpendicular to the rod, at distance d from the centre, follows directly from Eq. 2.4 and the Parallel Axis Theorem:

Formula 2.7

Straight uniform rod – general perpendicular axis at distance d from centre

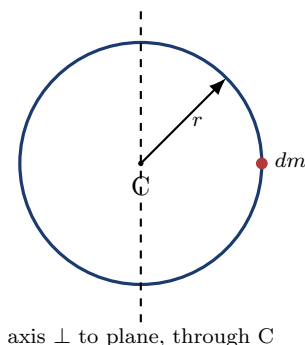
$$I(d) = \frac{ML^2}{12} + Md^2 \quad \left(0 \leq d \leq \frac{L}{2} \text{ on rod; } d > \frac{L}{2} \text{ if axis is outside the rod} \right)$$

About the length axis of the rod itself (axis coincides with the rod): $I = 0$ (thin rod, all mass on the axis).

Conditions This is only valid for a *straight* rod. A rod bent into an arc/circle no longer has this profile – every element then lies at the *same* distance r from the centre for a central axis, giving $I_C = mr^2$ (identical form to a ring, see Sec. 6.3) instead of $ML^2/12$.

5 Moment of Inertia of Ring, Disc and Sector of a Disc

5.1 Uniform Ring (circular hoop) — about the Central Axis ($\perp$ to plane)



Derivation. Every mass element dm of a uniform ring of radius r lies at the *same* perpendicular distance r from the central axis (perpendicular to the plane of the ring). Hence r can be pulled out of the integral:

$$I = \int r^2 dm = r^2 \int dm = r^2 M$$

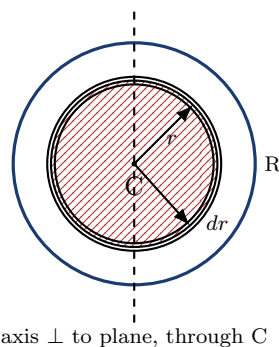
Formula 2.8

M.I. of a uniform ring about the central axis $\perp$ to its plane

$$I_C = Mr^2$$

Conditions Holds even if the mass is *non-uniformly* distributed around the ring, since every element is still at distance r from the axis; only $\int dm = M$ is used, not the distribution. Same axis also equals the M.I. of a thin rod bent into a circular arc of radius r (every arc element again at distance r): $I_C = mr^2$.

5.2 Uniform Disc — about the Central Axis ($\perp$ to plane)



Derivation. A uniform disc of mass M , radius R has surface density $\sigma = \frac{M}{\pi R^2}$. Slice it into thin concentric rings of radius r , width dr ; area of this elemental ring $= 2\pi r dr$, so $dm = \sigma (2\pi r dr)$. Every point of this elemental ring is at distance r from the axis, so by Eq. 2.8, $dI = r^2 dm$:

$$I = \int_0^R r^2 \sigma 2\pi r dr = 2\pi \sigma \int_0^R r^3 dr = 2\pi \sigma \frac{R^4}{4} = 2\pi \left(\frac{M}{\pi R^2} \right) \frac{R^4}{4}$$

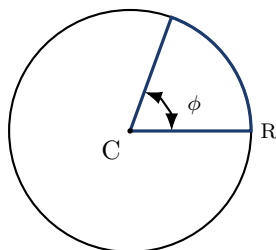
Formula 2.9

M.I. of a uniform disc about the central axis $\perp$ to its plane

$$I_C = \frac{MR^2}{2}$$

Conditions Disc is thin, uniform, planar (thickness negligible). By the Perpendicular Axis Theorem (Sec. 7), M.I. about any diameter is $I_{diam} = \frac{1}{4}MR^2$ (derived explicitly in Sec. 7).

5.3 Sector of a Uniform Disc — about the Central Axis ($\perp$ to plane, through the disc's centre)



Derivation. Consider a sector of a uniform disc subtending angle ϕ at the centre, mass m , radius r . Because the disc is uniform, the surface density σ of the sector is identical to that of the full disc of the same radius: $\sigma = \frac{m}{\frac{1}{2}\phi r^2} = \frac{M_{full\ disc}}{\pi r^2}$. Slicing the sector into elemental rings (radius x , thickness dx , but spanning only angle ϕ) gives element mass $dm = \sigma (x \phi dx)$, again with every point of the strip at distance x from the axis:

$$I = \int_0^r x^2 \sigma x \phi dx = \sigma \phi \int_0^r x^3 dx = \sigma \phi \frac{r^4}{4} = \frac{m}{\frac{1}{2}\phi r^2} \cdot \phi \cdot \frac{r^4}{4}$$

Formula 2.10

M.I. of a sector of a uniform disc, about the central axis $\perp$ to plane

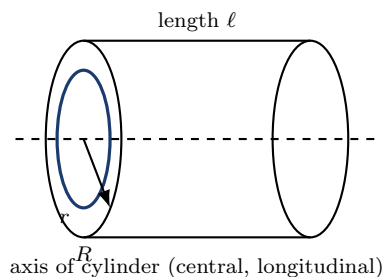
$$I_C = \frac{mr^2}{2}$$

Conditions Remarkably *independent of the sector angle ϕ* – identical in form to the full-disc formula, because M.I. per unit mass about this axis depends only on the radial mass distribution, which is unchanged by removing an angular slice (a uniform disc is “radially self-similar”). Axis must pass through the centre *of the disc* (the sector’s vertex), not through the sector’s own centroid.

Common Trap Do not confuse the sector’s own centre of mass (which lies along the bisector of ϕ , at radial distance $\frac{4r \sin(\phi/2)}{3\phi}$ from the vertex) with the disc’s geometric centre C used as the axis above. If a problem asks for M.I. about an axis through the *sector’s own centroid*, the Parallel Axis Theorem must be applied using this centroid distance.

6 Moment of Inertia of Cylinders

6.1 Homogeneous Solid Cylinder — about its own (Central Longitudinal) Axis



Derivation. A solid cylinder of mass M , radius R , length ℓ can be sliced into thin coaxial cylindrical *shells* of radius r , thickness dr . Each such shell is effectively a ring of mass dm at distance r from the axis, so $dI = r^2 dm$, exactly the disc derivation of Sec. 6.2 (the length ℓ plays no role since every shell at radius r , regardless of its position along the axis, is at the same perpendicular distance r):

$$dm = \rho(2\pi r dr \ell), \quad I = \int_0^R r^2 \rho 2\pi r \ell dr = 2\pi \rho \ell \frac{R^4}{4} = \frac{1}{2} (\rho \pi R^2 \ell) R^2 = \frac{1}{2} MR^2$$

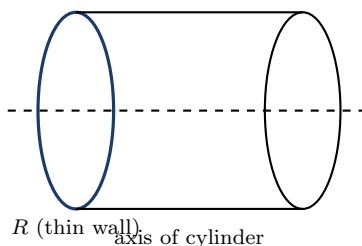
Formula 2.11

M.I. of a homogeneous solid cylinder about its own (central longitudinal) axis

$$I_C = \frac{MR^2}{2}$$

Conditions Identical in form to a disc (Eq. 2.9) because M.I. about the longitudinal axis is purely a function of the radial mass distribution – the length ℓ is irrelevant, so a solid cylinder is simply an infinite stack of discs about the same axis. *Does not apply* to M.I. about a diameter (transverse axis) through the centre – that combines both the "disc" term and a "rod-like" length term (a standard added result: $I_{diam} = M\left(\frac{R^2}{4} + \frac{\ell^2}{12}\right)$, obtained by superposing thin-disc contributions using the parallel-axis theorem along the length).

6.2 Hollow (Thin-Walled) Cylinder — about its own Axis



Derivation. For a thin-walled hollow cylinder (all mass on the curved surface at radius R), every mass element is at the same distance R from the axis, exactly as for a ring:

$$I = \int R^2 dm = R^2 \int dm = MR^2$$

Formula 2.12

M.I. of a thin hollow cylinder (or cylindrical shell) about its axis

$$I_C = MR^2$$

Thick hollow cylinder (annular / pipe), inner radius R_1 , outer radius R_2 :

Derivation. Treat as a solid cylinder of outer radius R_2 minus a solid cylinder of inner radius R_1 , of the *same density* ρ and length ℓ , so that the difference in masses reproduces the annulus. Using Eq. 2.11 for each and mass $\propto R^2$ for a given ρ, ℓ :

$$I = \frac{1}{2}M_2R_2^2 - \frac{1}{2}M_1R_1^2, \quad M_1 = \rho\pi R_1^2\ell, \quad M_2 = \rho\pi R_2^2\ell \Rightarrow I = \frac{1}{2}\rho\pi\ell (R_2^4 - R_1^4)$$

With total mass of the annulus $M = \rho\pi\ell(R_2^2 - R_1^2)$, and using $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$:

Formula 2.13

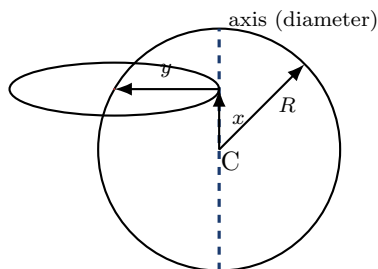
M.I. of a thick hollow cylinder (annulus), inner radius R_1 , outer radius R_2

$$I_C = \frac{M(R_1^2 + R_2^2)}{2}$$

Conditions Reduces to Eq. 2.11 (solid cylinder) when $R_1 \rightarrow 0$, and to Eq. 2.12 (thin shell) when $R_1 \rightarrow R_2 = R$. Always check which of these two hollow-cylinder formulas (Eq. 2.12 thin-walled vs Eq. 2.13 thick-walled/annular) the question intends – “hollow cylinder” without further data almost always means the thin-walled shell, Eq. 2.12.

7 Moment of Inertia of Spheres

7.1 Solid Sphere — about a Diameter



Derivation. Slice the solid sphere (mass M , radius R , density $\rho = \frac{M}{\frac{4}{3}\pi R^3}$) into thin circular discs perpendicular to the chosen diametral axis. A disc at height x from the centre has radius $y = \sqrt{R^2 - x^2}$, thickness dx , mass $dm = \rho \pi y^2 dx$. Using the disc formula (Eq. 2.9) for the M.I. of *this elemental disc about the same central axis*, $dI = \frac{1}{2} dm y^2$:

$$\begin{aligned} dI &= \frac{1}{2} \rho \pi y^4 dx = \frac{1}{2} \rho \pi (R^2 - x^2)^2 dx \\ I &= \frac{1}{2} \rho \pi \int_{-R}^R (R^2 - x^2)^2 dx = \frac{1}{2} \rho \pi \int_{-R}^R (R^4 - 2R^2 x^2 + x^4) dx = \frac{1}{2} \rho \pi \left[R^4(2R) - 2R^2 \cdot \frac{2R^3}{3} + \frac{2R^5}{5} \right] \\ &= \frac{1}{2} \rho \pi \left(2R^5 - \frac{4}{3} R^5 + \frac{2}{5} R^5 \right) = \frac{1}{2} \rho \pi \left(\frac{8}{15} \right) 2R^5 = \frac{8}{15} \pi \rho R^5 \end{aligned}$$

Substituting $\rho = \frac{3M}{4\pi R^3}$:

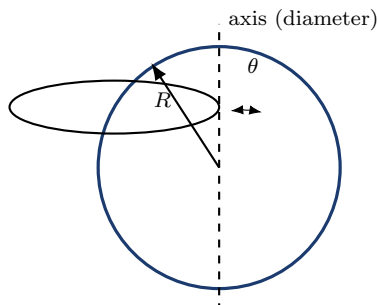
Formula 2.14

M.I. of a uniform solid sphere about a diameter

$$I_C = \frac{2}{5} MR^2$$

Conditions Sphere must be uniform (or, more generally, made of concentric uniform-density shells – “radially stratified”) for this simple coefficient; a non-uniform density $\rho(r)$ changes the numerical factor and must be re-derived. Applies to *any* diameter, since a uniform sphere is symmetric.

7.2 Hollow Sphere (Thin Spherical Shell) — about a Diameter



Derivation. Consider a thin uniform spherical shell of mass m , radius R , surface density $\sigma = \frac{m}{4\pi R^2}$. Slice it into elemental *rings* using the polar angle θ measured from the axis: a ring at angle θ has radius $y = R \sin \theta$, width $R d\theta$, so its area is $dA = 2\pi R \sin \theta \cdot R d\theta$, and its mass is $dm = \sigma 2\pi R^2 \sin \theta d\theta$. Every point of this ring is at distance $y = R \sin \theta$ from the axis, so by Eq. 2.8, $dI = y^2 dm$:

$$dI = (R \sin \theta)^2 \cdot \sigma 2\pi R^2 \sin \theta d\theta = 2\pi \sigma R^4 \sin^3 \theta d\theta$$

$$I = 2\pi\sigma R^4 \int_0^\pi \sin^3 \theta d\theta = 2\pi\sigma R^4 \cdot \frac{4}{3} = \frac{8}{3}\pi\sigma R^4$$

Substituting $\sigma = \frac{m}{4\pi R^2}$:

Formula 2.15

M.I. of a thin hollow sphere (spherical shell) about a diameter

$$I_C = \frac{2}{3}mR^2$$

Conditions All mass lies on the surface at radius R (shell thickness $\ll R$). Same coefficient regardless of where the axis is drawn, as long as it is a diameter, by spherical symmetry.

Bonus – thick hollow sphere (spherical shell with finite wall), inner radius r , outer radius R :

Derivation. Treat as a solid sphere of radius R minus a concentric solid sphere of radius r , both of the same density ρ , using Eq. 2.14: $I = \frac{2}{5}\rho \cdot \frac{4}{3}\pi R^5 - \frac{2}{5}\rho \cdot \frac{4}{3}\pi r^5 = \frac{8}{15}\pi\rho (R^5 - r^5)$, and with total mass $m = \frac{4}{3}\pi\rho(R^3 - r^3)$:

Formula 2.16

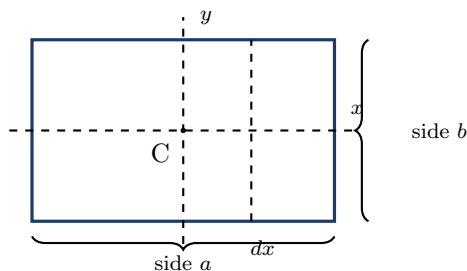
M.I. of a thick-walled hollow sphere (spherical shell), inner radius r , outer radius R

$$I_C = \frac{2}{5}m \left(\frac{R^5 - r^5}{R^3 - r^3} \right)$$

Conditions Reduces to Eq. 2.14 as $r \rightarrow 0$, and to Eq. 2.15 as $r \rightarrow R$ (using L'Hopital / binomial expansion for the thin-shell limit).

8 Moment of Inertia of Plates

8.1 Rectangular Plate



Derivation. Let the uniform rectangular plate have mass M , sides a (along x) and b (along y), centre at origin, surface density $\sigma = \frac{M}{ab}$.

(i) **In-plane axis along y (i.e. x measured across width a):** slice into strips parallel to the y -axis, each of width dx , length b , at distance x from the axis: $dm = \sigma b dx$.

$$I_y = \int_{-a/2}^{a/2} x^2 \sigma b dx = \sigma b \cdot \frac{2}{3} \left(\frac{a}{2} \right)^3 = \frac{\sigma a^3 b}{12} = \frac{Ma^2}{12}$$

By the identical argument (interchanging $a \leftrightarrow b$), $I_x = \frac{Mb^2}{12}$.

(ii) **Perpendicular axis through centre** follows immediately from the Perpendicular Axis Theorem (Sec. 7): $I_z = I_x + I_y$.

Formula 2.17

M.I. of a uniform rectangular plate, sides $a \times b$, mass M

$$I_x = \frac{Mb^2}{12} \text{ (in-plane, along length } a) \quad I_y = \frac{Ma^2}{12} \text{ (in-plane, along length } b)$$

$$I_C (\perp \text{ to plane, through centre}) = \frac{M(a^2 + b^2)}{12}$$

Conditions x, y axes must pass through the centroid and be parallel to the sides; the z -formula requires all three axes to be concurrent at the centre (Perpendicular Axis Theorem applies only to plane laminae). For a square plate ($a = b = s$): $I_C = \frac{Ms^2}{6}$.

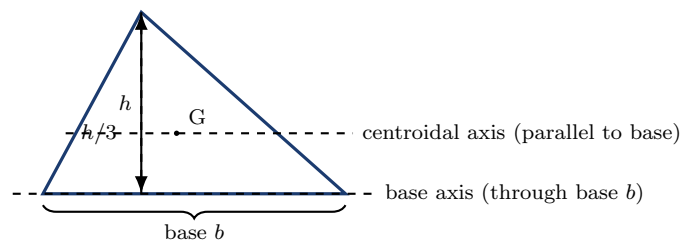
Formula 2.18

M.I. of a rectangular plate about an edge (parallel to the other side)

$$I_{edge} = I_x + M \left(\frac{b}{2} \right)^2 = \frac{Mb^2}{12} + \frac{Mb^2}{4} = \frac{Mb^2}{3}$$

Conditions Obtained directly by the Parallel Axis Theorem, shifting the centroidal in-plane axis by $b/2$ to lie along the edge.

8.2 Triangular Plate



Derivation. Let the uniform triangular lamina have mass M , base b , height h (apex at any horizontal position – the derivation below is independent of that position, since a horizontal shear does not change the width of a strip at a given height). Surface density $\sigma = \frac{M}{\frac{1}{2}bh} = \frac{2M}{bh}$.

Take the axis *along the base*. A horizontal strip at height y above the base ($0 \leq y \leq h$) has width $w(y) = b \left(1 - \frac{y}{h}\right)$ (linear taper to zero at the apex) and mass $dm = \sigma w(y) dy$:

$$I_{base} = \int_0^h y^2 \sigma b \left(1 - \frac{y}{h}\right) dy = \sigma b \int_0^h \left(y^2 - \frac{y^3}{h}\right) dy = \sigma b \left(\frac{h^3}{3} - \frac{h^3}{4}\right) = \frac{\sigma b h^3}{12} = \frac{2M}{bh} \cdot \frac{bh^3}{12}$$

Formula 2.19

M.I. of a uniform triangular plate about an axis *along its base*

$$I_{base} = \frac{Mh^2}{6}$$

Conditions h is the perpendicular height from the base to the opposite vertex; the axis is the base line itself, lying in the plane of the lamina.

Derivation. The centroid G of any triangle lies at height $h/3$ from the base (standard geometric fact – the median is divided 2 : 1 by the centroid). Apply the Parallel Axis Theorem *in reverse*, treating the base-axis as the “parallel” axis and the centroidal axis as the reference axis:

$$I_{base} = I_G + M \left(\frac{h}{3}\right)^2 \Rightarrow I_G = \frac{Mh^2}{6} - \frac{Mh^2}{9} = \frac{Mh^2(3-2)}{18}$$

Formula 2.20

M.I. of a uniform triangular plate about a centroidal axis parallel to the base

$$I_G = \frac{Mh^2}{18}$$

Conditions Axis passes through the centroid and is parallel to the base used to define h ; a different choice of base (and its own height) gives a different I_G for a scalene triangle – always identify *which* side is being used as the base.

Formula 2.21

M.I. of a uniform triangular plate about a perpendicular axis through the centroid (through the plane)

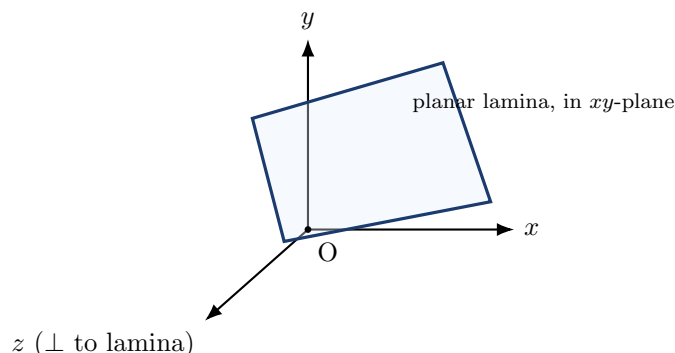
$$I_{G,\perp} = \frac{M(a^2 + b^2 + c^2)}{36}$$

Conditions Here a, b, c are the lengths of the three sides of the triangle. This is a standard identity for a uniform triangular lamina (the mass analogue of the area polar moment of a triangle about its centroid); it can be verified against Eq. 2.19–2.20 for the special case of an equilateral triangle of side s (where $h = \frac{\sqrt{3}}{2}s$): both routes give $I_{G,\perp} = \frac{Ms^2}{12}$.

Common Trap For a *right-angled* triangular plate with legs p (base) and q (height), the perpendicular-through-centroid result Eq. 2.21 still applies with $a = p$, $b = q$, $c = \sqrt{p^2 + q^2}$ (hypotenuse) – students often mistakenly try to use the rectangular-plate formula $M(a^2 + b^2)/12$, which is wrong for a triangle by both the coefficient (12 vs 36) and the physical shape.

9 Theorems on Moment of Inertia

9.1 Theorem of Perpendicular Axes



Derivation. Consider a planar lamina lying entirely in the xy -plane, with O the origin lying in that plane. Every mass element dm at (x, y) has distance $\sqrt{x^2 + y^2}$ from the z -axis (perpendicular to the lamina through O), distance $|y|$ from the x -axis, and distance $|x|$ from the y -axis.

$$I_z = \int (x^2 + y^2) dm = \int y^2 dm + \int x^2 dm = I_x + I_y$$

Formula 3.1

Theorem of Perpendicular Axes

$$I_z = I_x + I_y$$

Conditions Applies *only* to a plane lamina (a body that lies entirely in a plane, including a flat lamina or a rod/wire bent into a plane curve) – never to a 3-D body like a solid sphere or solid cylinder. The three axes Ox, Oy, Oz must be mutually perpendicular and **concurrent** (all pass through the same point O , which must itself lie in the plane of the lamina), with Oz perpendicular to the lamina and Ox, Oy lying in its plane.

Worked application (disc about a diameter): for a disc, symmetry gives $I_x = I_y$ (both diameters). Since $I_z = \frac{1}{2}MR^2$ (Eq. 2.9):

$$I_z = 2I_x \Rightarrow I_{diam} = I_x = \frac{1}{2}I_z = \frac{1}{4}MR^2$$

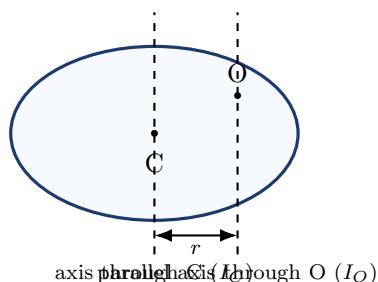
Formula 3.2

M.I. of a uniform disc / ring about a diameter (in-plane centroidal axis)

$$I_{diam, disc} = \frac{MR^2}{4}, \quad I_{diam, ring} = \frac{MR^2}{2}$$

Conditions Ring value follows the same route: $I_z = MR^2$ and $I_x = I_y$ by symmetry $\Rightarrow I_{diam} = \frac{1}{2}MR^2$.

9.2 Theorem of Parallel Axes (Steiner's Theorem)



Derivation. Let a rigid body of mass M have its centre of mass at C . Let I_C be its M.I. about an axis through C , and I_O its M.I. about a *parallel* axis through any other point O , at perpendicular distance r from the axis through C . Place the axis through C along the z -direction, with C at the origin of the xy -plane and O at $(r, 0)$. A mass element at (x, y) is at distance $\sqrt{x^2 + y^2}$ from the C -axis and $\sqrt{(x-r)^2 + y^2}$ from the O -axis:

$$I_O = \int [(x-r)^2 + y^2] dm = \int (x^2 + y^2) dm - 2r \int x dm + r^2 \int dm$$

Since C is the centre of mass, $\int x dm = 0$ (by definition of centre of mass measured from itself). Hence:

$$I_O = I_C + Mr^2$$

Formula 3.3

Theorem of Parallel Axes (Steiner's Theorem)

$$I_O = I_C + Mr^2$$

Conditions The reference axis *through C* must pass through the **centre of mass** – the theorem does *not* hold between two arbitrary parallel axes neither of which passes through C . r is the perpendicular distance between the two parallel axes. It follows that, among all parallel axes, M.I. is **minimum** about the axis through the centre of mass.

Key Points The correction term Mr^2 is exactly the M.I. of a point particle of mass M (the whole body's mass) situated at the centre of mass, about the new axis. This reflects that general plane motion is always separable into (i) translation of the centre of mass and (ii) rotation about the centre of mass – the same split used throughout Part IV.

Common Trap Parallel Axis Theorem *adds* to I_C ($I_O = I_C + Mr^2$, always $I_O \geq I_C$); Perpendicular Axis Theorem *splits* I_z into in-plane components ($I_z = I_x + I_y$). They solve different problems and are frequently combined in sequence, e.g. disc about a tangent in its own plane: $I_{\text{tangent}} = I_{\text{diam}} + MR^2 = \frac{1}{4}MR^2 + MR^2 = \frac{5}{4}MR^2$.

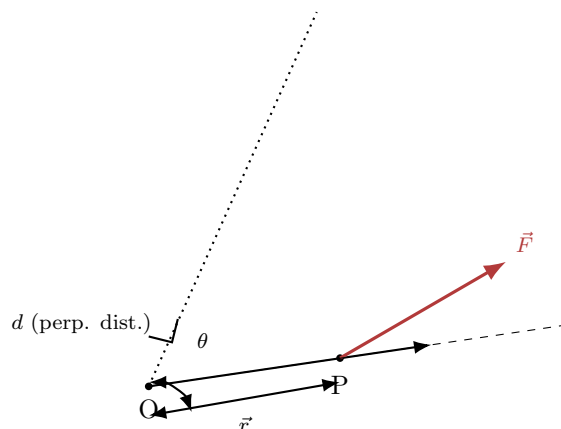
9.3 Master Table – Moment of Inertia of Standard Uniform Bodies

Body (mass M)	Axis	Moment of Inertia
Uniform rod, length L	centre, $\perp$ to rod	$\frac{ML^2}{12}$
Uniform rod, length L	one end, $\perp$ to rod	$\frac{ML^2}{3}$
Uniform rod bent into an arc, radius r	centre of curvature, $\perp$ to plane of arc	Mr^2
Uniform ring, radius R	central, $\perp$ to plane	MR^2
Uniform ring, radius R	diameter	$\frac{MR^2}{2}$
Sector of a uniform disc, radius r	centre of disc, $\perp$ to plane	$\frac{mr^2}{2}$
Uniform disc, radius R	central, $\perp$ to plane	$\frac{MR^2}{2}$
Uniform disc, radius R	diameter	$\frac{MR^2}{4}$
Solid cylinder, radius R	own (longitudinal) axis	$\frac{MR^2}{2}$
Thin hollow cylinder, radius R	own axis	MR^2
Thick hollow cylinder, radii R_1, R_2	own axis	$\frac{M(R_1^2 + R_2^2)}{2}$
Solid sphere, radius R	diameter	$\frac{2}{5}MR^2$
Thin hollow sphere (shell), radius R	diameter	$\frac{2}{3}MR^2$
Thick hollow sphere, radii r, R	diameter	$\frac{2}{5}M\left(\frac{R^5 - r^5}{R^3 - r^3}\right)$
Rectangular plate, sides $a \times b$	$\perp$ to plane, through centre	$\frac{M(a^2 + b^2)}{12}$
Rectangular plate, sides $a \times b$	in-plane centroidal, along side a	$\frac{Mb^2}{12}$
Triangular plate, base b , height h	along the base	$\frac{Mh^2}{6}$
Triangular plate, base b , height h	centroidal, $\parallel$ base	$\frac{Mh^2}{18}$
Triangular plate, sides a, b, c	$\perp$ to plane, through centroid	$\frac{M(a^2 + b^2 + c^2)}{36}$

Part III — Torque

10 Torque About a Point

10.1 Definition



Derivation. Torque (moment of force) is the rotational analogue of force. For a force $\vec{F}$ applied at point P , with position vector $\vec{r}$ drawn from the point O about which torque is being computed to P , the torque is defined as a cross product:

Formula 4.1

Torque about a point O

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad |\vec{\tau}| = rF \sin \theta = F(r \sin \theta) = F \cdot d$$

Conditions θ is the angle between $\vec{r}$ and $\vec{F}$. $d = r \sin \theta$ is the **perpendicular distance** from O to the **line of action** of $\vec{F}$ (the “moment arm”); this is often the fastest way to compute magnitude without resolving vectors. Direction of $\vec{\tau}$ is given by the right-hand rule, perpendicular to the plane containing $\vec{r}$ and $\vec{F}$. SI unit: N m (not to be confused with Joule – torque is not a form of energy, so the unit is never written as J).

Common Trap Torque depends on the choice of reference point O (unlike force). Always state / fix the point (or axis) about which torque is being calculated before writing any equation, and use the *same* point consistently for every force in the problem.

11 Torque About an Axis

For rotation of a rigid body about a *fixed axis* (not just a point), only the component of $\vec{\tau}$ (computed about any point on the axis) *along* the axis actually produces angular acceleration; components perpendicular to the axis are cancelled by the axis’s bearing/support reactions.

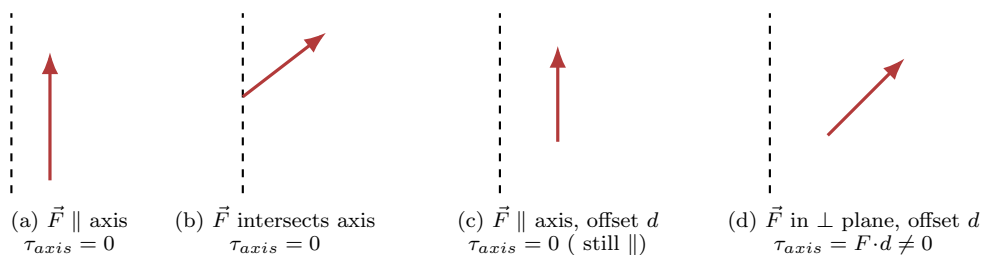
Formula 4.2

Torque about a fixed axis

$$\tau_{axis} = \vec{\tau} \cdot \hat{n} = (\vec{r} \times \vec{F}) \cdot \hat{n}$$

Conditions $\hat{n}$ is the unit vector along the axis. Equivalent practical rule: resolve $\vec{F}$ into a component $F_{\parallel}$ parallel to the axis (produces no torque about the axis) and a component $F_{\perp}$ in the plane perpendicular to the axis; then $\tau_{axis} = F_{\perp} \cdot d$, where d is the perpendicular distance from the axis to the line of action of $F_{\perp}$, within that plane.

Four standard cases for the line of action of $\vec{F}$ relative to the axis:



Conditions Only case (d) – force lying in (or having a component in) a plane perpendicular to the axis, *and* not intersecting the axis – produces a non-zero torque about that axis. Forces parallel to the axis, or whose line of action intersects/crosses the axis, contribute zero torque about that axis (though they may contribute torque about a *different* point/axis).

12 The Rotational Equation of Motion: $\tau = I\alpha$

Derivation. For a rigid body rotating about a fixed axis, each particle i of mass m_i at distance r_i from the axis experiences a tangential force $F_{t,i} = m_i a_{t,i} = m_i r_i \alpha$ (Eq. 1.3) from the net effect of all real forces on it. The torque of this tangential force about the axis is $\tau_i = F_{t,i} r_i = m_i r_i^2 \alpha$. Summing over all particles (internal action-reaction torque pairs cancel by Newton's third law, since internal forces act along the line joining particles):

$$\tau_{net} = \sum_i m_i r_i^2 \alpha = \left(\sum_i m_i r_i^2 \right) \alpha$$

Formula 4.3

Newton's Second Law for Rotation

$$\vec{\tau}_{net} = I\vec{\alpha} \quad (\text{about a fixed axis, or about the centre of mass for general plane motion})$$

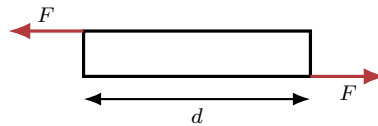
Conditions I is the M.I. about the *same* axis about which τ_{net} is computed. α is the same for every particle only when the axis is fixed in direction (rigid body assumption). For general plane motion (translating + rotating axis), $\tau_{cm} = I_{cm}\alpha$ still holds *if* torques are computed about the centre of mass – this is a special, very useful result that holds even though the CM may itself be accelerating.

12.1 How to Set Up the Torque Equation – Step by Step

When & How to Use

1. **Choose the axis.** Fixed-axis problem $\Rightarrow$ use the actual (given) axis. Free rigid body (rolling, falling rod, etc.) $\Rightarrow$ usually choose the centre of mass, or the ICR (Sec. 11) if it simplifies the algebra (no unknown friction/normal-force torque appears about it).
2. **Draw the free body diagram** of the rigid body alone, marking every external force at its actual point of application (position matters for torque, unlike for $F = ma$).
3. **Assign a positive sense** (say CCW +ve) and stick to it for every torque and for α .
4. **Compute each torque** as $F \times$ (perpendicular distance from axis to line of action), with sign according to step 3.
5. **Write $\sum \tau = I\alpha$** about the chosen axis, and, if the axis is accelerating (translating), also write $F_{net} = Ma_{cm}$ for the centre of mass – two equations for two unknowns is standard for rolling/general plane motion problems.
6. **Constraint relation:** if there is rolling without slipping or a string over a pulley, connect a_{cm} and α (e.g. $a_{cm} = R\alpha$) *before* solving.

13 Couple



A **couple** is a pair of equal, opposite and *parallel* (non-collinear) forces. A couple produces pure rotation (zero net force, non-zero net torque), and its torque is the *same about every point* in space.

Formula 4.4

Torque of a couple

$$\tau_{\text{couple}} = F \cdot d$$

Conditions d is the perpendicular distance between the two lines of action (the “arm” of the couple). Because $F_{\text{net}} = 0$, a couple alone cannot accelerate the centre of mass – it only changes ω .

Key Points Torque is additive: net torque about any point/axis is the vector (or, for a fixed axis, signed scalar) sum of the torques of all external forces about that same point/axis. Static equilibrium of a rigid body requires *both* $\vec{F}_{\text{net}} = 0$ and $\vec{\tau}_{\text{net}} = 0$ about *any* one point (if $\tau_{\text{net}} = 0$ about one point and $F_{\text{net}} = 0$, then $\tau_{\text{net}} = 0$ about *every* point).

Part IV — Combined Translational and Rotational Motion

14 General Plane Motion: Decomposition

Derivation. The position of any particle P of a rigid body can be written as $\vec{r}_P = \vec{r}_{cm} + \vec{r}_{P/cm}$, where $\vec{r}_{P/cm}$ is the position of P relative to the centre of mass. Differentiating (and using $|\vec{r}_{P/cm}| = \text{const}$ for a rigid body, so its only possible change is rotation, $\vec{v}_{P/cm} = \vec{\omega} \times \vec{r}_{P/cm}$):

Formula 5.1

Velocity and acceleration for general plane motion (translation + rotation about CM)

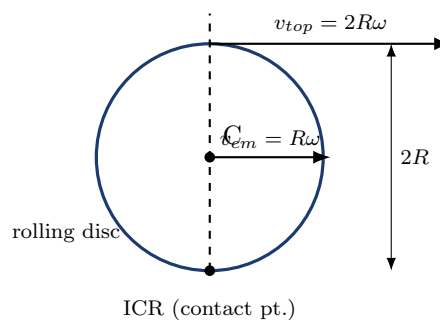
$$\vec{v}_P = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_{P/cm} \quad \vec{a}_P = \vec{a}_{cm} + \vec{\alpha} \times \vec{r}_{P/cm} - \omega^2 \vec{r}_{P/cm}$$

Conditions $\vec{\omega}, \vec{\alpha}$ are the same for every point of the rigid body at a given instant (that is what "rigid" means). This decomposition – translation of the CM *plus* rotation about the CM – is completely general and is the basis of every rolling/toppling/general-motion problem.

15 Instantaneous Axis of Rotation (IAR) and Instantaneous Centre of Rotation (ICR)

15.1 Definition

At any instant, the general plane motion of a rigid lamina is equivalent to a *pure rotation* (with the same ω) about some unique point in the plane of motion (or line, in 3-D) at which the instantaneous velocity is zero. This point is the **Instantaneous Centre of Rotation (ICR)**; the corresponding line perpendicular to the plane through this point is the **Instantaneous Axis of Rotation (IAR)**.



15.2 Locating the ICR – Three Standard Methods

When & How to Use

- Known contact / constraint point:** for a body rolling without slipping on a stationary surface, the contact point has zero velocity by definition of "no slipping" – it *is* the ICR (as in the rolling wheel above).
- Two velocity directions known:** draw perpendiculars to the velocity vectors of two points of the body; ICR is their intersection (velocity of any point is always perpendicular to the line joining it to the ICR, since the body instantaneously "rotates" about the ICR).
- Velocity magnitudes of two points on a line known:** the ICR lies on the line joining them (extended if necessary), located such that velocities are proportional to distance from the ICR ($v_1/r_1 = v_2/r_2 = \omega$).

Formula 5.2

Velocity of any point via the ICR

$$v_P = \omega \cdot r_{P,ICR}$$

Conditions $r_{P,ICR}$ is the distance of point P from the ICR, *at that instant only* – the ICR is generally a different geometric point of the body (or of space) at each successive instant, so this formula **cannot** be differentiated directly to get acceleration ($a_P \neq \alpha r_{P,ICR}$ in general, because the ICR itself has an acceleration). Acceleration must instead be found via Eq. 5.1 about the centre of mass, or via the general two-point acceleration relation.

Common Trap The ICR is, in general, a *geometrical* point that may not even be a material point of the body (e.g. a rolling wheel's ICR is on the ground, not inside the wheel's own material – though at the instant of contact it coincides with the contact particle of the wheel, which momentarily has zero velocity). Never differentiate $v = \omega r_{ICR}$ to find acceleration of the ICR-coincident point – the widely-used shortcut “acceleration of contact point of a rolling body = $\omega^2 R$ directed towards the centre” is a *separate*, correctly-derived result (Sec. 12.2), not a blind derivative of Eq. 5.2.

16 Rolling Motion

16.1 Condition for Pure Rolling (Rolling without Slipping)

Derivation. A round body (disc, ring, cylinder, sphere) of radius R moves on a surface with its centre having velocity v_{cm} and the body spinning with angular velocity ω . The velocity of the contact point P (bottom-most point) relative to the ground is the vector sum of the translational velocity of the centre and the rotational velocity of P about the centre:

$$v_P = v_{cm} - \omega R \quad (\text{taking forward translation and forward-rolling spin as positive, contact point velocity due to rotation is backward})$$

"Rolling without slipping" is defined as the contact point having *zero relative velocity* with respect to the surface (no sliding) at every instant, i.e. $v_P = 0$:

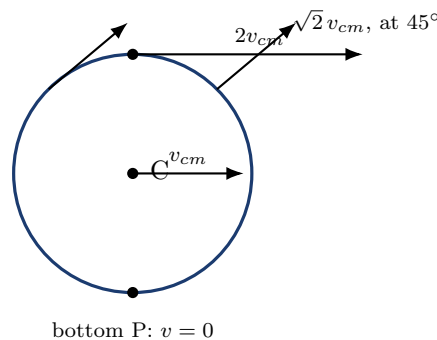
Formula 5.3

Condition for Pure Rolling

$$v_{cm} = \omega R \quad a_{cm} = \alpha R \quad (\text{differentiating, valid only if } R = \text{const})$$

Conditions Holds only when there is *no slipping* at the contact point (static friction, not kinetic, acts there, and it can take any value up to $\mu_s N$ as required by the dynamics – it is a constraint force, not necessarily at its maximum). If the body slips, $v_{cm} \neq \omega R$ and *kinetic* friction $= \mu_k N$ acts, until slipping stops.

16.2 Velocity of Points on a Rolling Body



Derivation. Using Eq. 5.1 with $\vec{v}_{cm}$ horizontal (magnitude $v = \omega R$), for a general point at angular position θ from the bottom (measured from the centre), position relative to centre $\vec{r}_{P/cm} = R(\sin \theta, -\cos \theta)$ so $\vec{\omega} \times \vec{r}_{P/cm}$ contributes a rotational velocity of magnitude $\omega R = v$ tangential to the rim. Vector addition of the uniform translational velocity v (forward) and this rotational velocity (tangential, magnitude v) at each point gives the results below; equivalently, using the ICR at the bottom-most point P (Eq. 5.2), $v_{point} = \omega \times (\text{distance from P})$.

Formula 5.4

Velocity of standard points of a rolling body (ICR at bottom contact point)

$$v_{bottom} = 0$$

$$v_{centre} = v_{cm} = \omega R$$

$$v_{top} = 2v_{cm} = 2\omega R$$

$$v_{(\text{at angle } \theta \text{ from bottom, on rim})} = 2v_{cm} \sin(\theta/2), \text{ directed perpendicular to the line joining that point to P}$$

Conditions Applies to a rigid, circular, rolling body (disc/ring/sphere/cylinder) rolling without slipping on a flat surface, with the ICR at the contact point. The top point always has velocity exactly *double* v_{cm} , directed forward – a frequently tested and highly useful special result.

16.3 Acceleration of the Contact Point of a Rolling Body

Derivation. Although the contact point's *velocity* is instantaneously zero, its *acceleration* is not (Sec. 10.2 warns against differentiating $v = \omega r_{ICR}$ naively). Using Eq. 5.1 directly for the bottom point P, with $\vec{r}_{P/cm} = (0, -R)$:

$$\vec{a}_P = \vec{a}_{cm} + \vec{\alpha} \times \vec{r}_{P/cm} - \omega^2 \vec{r}_{P/cm}$$

For pure rolling, $a_{cm} = \alpha R$ (horizontal) and the term $\vec{\alpha} \times \vec{r}_{P/cm}$ (also horizontal, magnitude αR) exactly *cancels* the horizontal a_{cm} term (they point in opposite senses for the bottom point), leaving only the centripetal term:

Formula 5.5

Acceleration of the contact point of a body in pure rolling

$$a_{contact} = \omega^2 R \quad \text{directed vertically upward, towards the centre}$$

Conditions Valid strictly for *uniform* pure rolling on a straight, flat surface (so that the contact point is always the geometric bottom of the circle). This is why, for pure rolling, the contact point's horizontal acceleration is zero even while $a_{cm} \neq 0$ – a key idea used to identify whether a wheel is undergoing pure rolling versus rolling-with-slipping in a given snapshot.

16.4 Kinetic Energy of a Rolling Body

Derivation. General plane motion K.E. splits into translational K.E. of the CM plus rotational K.E. about the CM (a direct consequence of Eq. 5.1, since cross terms vanish when summed over the body about its own CM):

$$K = \underbrace{\frac{1}{2} M v_{cm}^2}_{\text{translation of CM}} + \underbrace{\frac{1}{2} I_{cm} \omega^2}_{\text{rotation about CM}}$$

For *pure rolling*, substitute $v_{cm} = \omega R$ and write $I_{cm} = M k^2$ (radius of gyration, Eq. 2.3):

$$K = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} M k^2 \left(\frac{v_{cm}}{R} \right)^2 = \frac{1}{2} M v_{cm}^2 \left(1 + \frac{k^2}{R^2} \right)$$

Formula 5.6

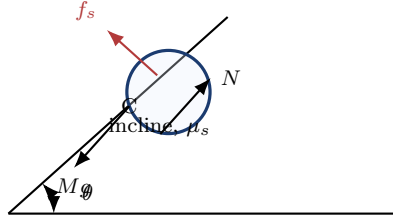
Kinetic energy of a body in pure rolling

$$K = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 = \frac{1}{2} M v_{cm}^2 \left(1 + \frac{k^2}{R^2} \right) = \frac{1}{2} I_{ICR} \omega^2$$

Conditions The last form ($\frac{1}{2} I_{ICR} \omega^2$) treats the instantaneous motion as *pure rotation* about the (instantaneous) ICR, using $I_{ICR} = I_{cm} + M R^2$ (Parallel Axis Theorem) – valid *only* for computing K.E. at that instant, since (as emphasised above) the ICR is not a fixed material point and this shortcut cannot be used for computing power ($= \tau \omega$ about a genuinely fixed axis) or angular momentum without care.

Key Points For pure rolling, the fraction of K.E. that is rotational is $\frac{k^2}{R^2 + k^2}$, and translational is $\frac{R^2}{R^2 + k^2}$. Ranking (largest k^2/R^2 = most rotational KE, slowest linear acceleration down an incline for a given M, R): hollow sphere > ring/hollow cylinder > disc/solid cylinder > solid sphere.

17 Rolling on an Inclined Plane



17.1 Dynamics (Force + Torque) Method

Derivation. A uniform round body (mass M , radius R , $I_{cm} = Mk^2$) rolls without slipping down a rough incline of angle θ . Forces on it: weight Mg (down), normal reaction N (perpendicular to incline), static friction f_s (up the incline, opposing the tendency to slip – direction confirmed by the algebra below).

Newton's second law along the incline (taking down-the-incline as positive):

$$Mg \sin \theta - f_s = Ma_{cm} \quad (i)$$

Torque equation about the centre of mass (friction is the only force with a non-zero moment arm R about C ; N and Mg pass through / act at C for a uniform symmetric body):

$$f_s R = I_{cm} \alpha = Mk^2 \alpha \quad (ii)$$

Rolling constraint: $a_{cm} = \alpha R \Rightarrow \alpha = a_{cm}/R$. Substitute into (ii): $f_s = \frac{Mk^2 a_{cm}}{R^2}$. Substitute into (i):

$$Mg \sin \theta - \frac{Mk^2 a_{cm}}{R^2} = Ma_{cm} \Rightarrow g \sin \theta = a_{cm} \left(1 + \frac{k^2}{R^2} \right)$$

Formula 5.7

Acceleration of a uniform body rolling without slipping down a rough incline

$$a_{cm} = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}} \quad \alpha = \frac{a_{cm}}{R} \quad f_s = \frac{Mg \sin \theta}{1 + \frac{k^2}{R^2}} = \frac{k^2}{k^2 + R^2} Mg \sin \theta$$

Conditions Rolling without slipping throughout (i.e. friction available is enough – see Sec. 12.3 for the exact requirement). a_{cm} is *independent* of M and R individually – it depends only on θ and the shape factor k^2/R^2 . Every body with the same k^2/R^2 (e.g. all solid spheres, of any mass or radius) accelerates identically down a given incline.

17.2 Energy Method (alternative derivation)

Derivation. Using energy conservation between the top (rest) and after descending a height $h = s \sin \theta$ along the incline (friction here is *static*, so it does *no* work – the contact point has zero instantaneous velocity, hence zero power dissipated):

$$Mgh = \frac{1}{2} M v_{cm}^2 \left(1 + \frac{k^2}{R^2} \right) \Rightarrow v_{cm} = \sqrt{\frac{2gh}{1 + k^2/R^2}} = \sqrt{\frac{2gs \sin \theta}{1 + k^2/R^2}}$$

Comparing with $v^2 = 2a_{cm}s$ (kinematics) reproduces Eq. 5.7 exactly.

Formula 5.8

Speed of a rolling body after descending height h on a rough (no-slip) incline

$$v_{cm} = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

Conditions Static friction does zero work in pure rolling, so mechanical energy is conserved even though friction is present – this is *not* true the moment the body slips (kinetic friction then dissipates energy, and this formula fails).

17.3 Minimum Coefficient of Friction Required for Pure Rolling

Derivation. From Eq. 5.7, $f_s = \frac{k^2}{k^2 + R^2} Mg \sin \theta$. For pure rolling to actually occur, the *available* static friction must be sufficient: $f_s \leq \mu_s N = \mu_s Mg \cos \theta$.

Formula 5.9

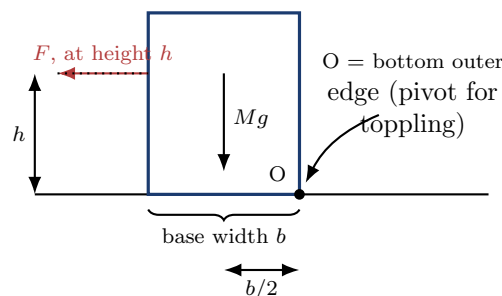
Condition on μ_s for pure rolling on an incline

$$\mu_s \geq \left(\frac{k^2}{k^2 + R^2} \right) \tan \theta$$

Conditions If the actual μ_s available is less than this, the body *slips while rolling*: kinetic friction $\mu_k Mg \cos \theta$ then acts (now doing negative work, dissipating energy), and $a_{cm} = g \sin \theta - \mu_k g \cos \theta$ (translational equation alone, since friction is now known exactly) together with $\alpha = \frac{f R}{I_{cm}} = \frac{\mu_k Mg R \cos \theta}{M k^2}$ (independent equations, no longer linked by $a = \alpha R$) must be used instead of Eq. 5.7.

Common Trap Static friction on a rolling body can point *either* up or down the incline (or be zero) depending on the problem (e.g. an accelerating cart carrying a rolling ball, or a yo-yo) – it is *not* always “up the incline, opposing sliding tendency due to gravity”. Always solve for its magnitude *and* sign from the equations; never assume its direction beforehand for problems other than the simple “released from rest on an incline” case above.

18 Toppling



Derivation. A block of mass M , base width b , height H_{cm} (height of the CM above the base), rests on a rough floor and is pushed by a horizontal force F applied at height h above the base. Two independent things can happen: the block can *slide* (if $F > \mu_s Mg$), or it can *topple* by rotating about its outer bottom edge O.

Toppling analysis: as F increases, the normal reaction N (which always equals Mg here, since there is no vertical acceleration) shifts *away* from F 's side to keep torque balance – take torques about the centre of mass: N acts at a horizontal distance x from the line of Mg (i.e. from the CM's vertical line) such that $Nx = Fh$ (rotational equilibrium, block still in equilibrium, not yet toppling), giving $x = \frac{Fh}{Mg}$.

The block can shift N 's line of action only until it reaches the outer edge O, at $x = b/2$ from the centre. **Toppling begins exactly when x reaches this maximum value $b/2$** (beyond this, N would have to act outside the base, which is impossible, so the block instead rotates about O):

$$\frac{Fh}{Mg} = \frac{b}{2} \Rightarrow F_{\text{topple}} = \frac{Mgb}{2h}$$

Formula 5.10

Force required to topple a block (about the bottom outer edge)

$$F_{\text{topple}} = \frac{Mgb}{2h}$$

Conditions b = base width (in the direction of tipping), h = height at which F is applied, M, g as usual. Assumes the block does not slide before this F is reached (see comparison below) and that F is horizontal.

Formula 5.11

Comparing sliding vs. toppling

$$F_{\text{slide}} = \mu_s Mg \qquad F_{\text{topple}} = \frac{Mgb}{2h}$$

$$\text{Block slides first if } \mu_s < \frac{b}{2h} ; \qquad \text{Block topples first if } \mu_s > \frac{b}{2h}$$

Conditions Whichever critical force is *smaller* occurs first as F is increased from zero. A tall, narrow block (small b/h) on a very rough surface (large μ_s) is prone to toppling before sliding; a short, wide block on a slippery surface slides before it can topple.

Formula 5.12

Toppling on a tilting/inclined surface (self-toppling under gravity alone)

$$\tan \theta_{\text{topple}} = \frac{b}{2H_{cm}}$$

Derivation. If instead of pushing the block, the supporting surface itself is tilted to angle θ , gravity's own line of action (through the CM, vertically down) moves toward the lower edge. Toppling begins when this vertical line through the CM just passes through the bottom edge O – i.e. when the horizontal offset of the CM from O, $= \frac{b}{2} \cos \theta - H_{cm} \sin \theta$ measured appropriately along the base, becomes zero, giving $\tan \theta_{\text{topple}} = \frac{b/2}{H_{cm}}$, where H_{cm} is measured from the base to the CM along the block.

Conditions Here too, compare with the sliding condition $\tan \theta_{\text{slide}} = \mu_s$: the block slides first if $\mu_s < \frac{b}{2H_{cm}}$, and topples first if $\mu_s > \frac{b}{2H_{cm}}$, exactly analogous to Eq. 5.11.

Common Trap Toppling is a question of **torque balance about the tipping edge**, not of the net force. A block can be in perfect translational equilibrium ($F_{\text{net}} = 0$) right up to the instant it topples – toppling is triggered purely by the normal reaction's line of action reaching the edge of the base, not by any force becoming “too large” in the Newton's-second-law sense.

Part V — Work, Energy and Angular Momentum in Rotation

19 Work and Power in Rotation

Derivation. For a rigid body rotating about a fixed axis under torque τ , work done by the torque as the body turns through $d\theta$ equals the work done by the corresponding tangential force through the arc length it moves: $dW = F_t(r d\theta) = (F_t r) d\theta = \tau d\theta$.

Formula 6.1

Work and Power of a Torque

$$dW = \tau d\theta \quad W = \int \tau d\theta \quad P = \frac{dW}{dt} = \tau \omega$$

Conditions τ is the net torque about the (fixed) axis of rotation; θ, ω are measured about the same axis. For constant τ : $W = \tau \Delta\theta$.

19.1 Work–Energy Theorem for Rotation

Formula 6.2

Rotational Work–Energy Theorem

$$W_{net} = \Delta K_{rot} = \frac{1}{2} I \omega_2^2 - \frac{1}{2} I \omega_1^2$$

Conditions I is about the fixed axis of rotation. For a body undergoing general plane motion, the *total* work–energy theorem uses total K.E. from Eq. 5.6: $W_{net} = \Delta (\frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2)$, with W_{net} computed as work done by *all* external forces through their actual (translational) displacements, plus, where applicable, work done by torques about the CM.

20 Angular Momentum

20.1 Angular Momentum of a Particle

Formula 6.3

Angular momentum of a particle about a point O

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}, \quad L = mvr \sin \theta = mv d$$

Conditions $\vec{r}$ is drawn from O to the particle; $d = r \sin \theta$ is the perpendicular distance from O to the line of $\vec{v}$ (moment-arm form, analogous to torque). Depends on the choice of O, exactly like torque.

20.2 Angular Momentum of a Rigid Body About a Fixed Axis

Derivation. Summing $L_i = m_i v_i r_i = m_i (\omega r_i) r_i = m_i r_i^2 \omega$ over all particles of a body rotating about a fixed axis (each particle's velocity is purely tangential, perpendicular to r_i , so $\sin \theta = 1$):

$$L = \sum_i m_i r_i^2 \omega = I \omega$$

Formula 6.4

Angular momentum about a fixed axis

$$L = I\omega$$

Conditions I and L are about the same fixed axis. Direction of $\vec{L}$ is along $\vec{\omega}$ (right-hand rule) for rotation about a principal/symmetry axis (true for all the symmetric standard bodies of Part II).

20.3 Angular Momentum in General Plane Motion

Formula 6.5

Angular momentum of a body in general plane motion, about a fixed point O

$$\vec{L}_O = I_{cm}\vec{\omega} + M(\vec{r}_{cm} \times \vec{v}_{cm})$$

Conditions First term = "spin" angular momentum about the CM; second term = "orbital" angular momentum of the CM (treated as a point mass M) about O. $\vec{r}_{cm}$ is drawn from O to the CM. For pure rolling about the contact point taken as O (instantaneously coincident with the ICR), this reduces neatly to $L = I_{cm}\omega + Mv_{cm}R = I_{cm}\omega + MR^2\omega = I_{ICR}\omega$ (consistent with the K.E. shortcut of Eq. 5.6).

20.4 Relation between Torque and Angular Momentum

Derivation. Differentiating $\vec{L} = \vec{r} \times \vec{p}$ with respect to time: $\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} = \vec{v} \times m\vec{v} + \vec{r} \times \vec{F} = 0 + \vec{\tau}$ (since $\vec{v} \times \vec{v} = 0$).

Formula 6.6

Rotational analogue of Newton's Second Law (general form)

$$\vec{\tau}_{net} = \frac{d\vec{L}}{dt}$$

Conditions This is the most general statement, valid about any *fixed* point O, or about the centre of mass (even if accelerating); it reduces to $\tau = I\alpha$ (Eq. 4.3) when I is constant (rigid body, fixed axis).

20.5 Conservation of Angular Momentum

Formula 6.7

Conservation of Angular Momentum

$$\text{If } \vec{\tau}_{net} = 0 \text{ about a point/axis, } \vec{L} = \text{constant} \Rightarrow I_1\omega_1 = I_2\omega_2$$

Conditions Applies about the *same* fixed point/axis throughout (or the CM). Commonly used when I of a system changes due to internal rearrangement of mass (ice-skater pulling arms in, person walking on a rotating disc, collision of a particle with a rotating rod) while external torque about the relevant axis is zero (e.g. gravity/normal reaction pass through or are parallel to the axis).

Key Points Linear momentum, angular momentum and energy are three separate conserved quantities – a problem may conserve one, two, or all three depending on the forces/torques present (e.g. a perfectly inelastic collision conserves linear and/or angular momentum but *not* kinetic energy). Always check independently: is $F_{net} = 0$? Is $\tau_{net} = 0$ about the axis of interest? Are all forces conservative / is there no friction doing work?

Part VI — Exam Tools

21 Master Formula Atlas

Family	Formula	Condition / Keyword
Ang. kinematics	$\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\theta$	α constant
Linear-angular	$v = \omega r$, $a_t = \alpha r$, $a_n = \omega^2 r$	fixed axis, particle at r
M.I. definition	$I = \int r^2 dm = \sum m_i r_i^2$	any axis
Radius of gyration	$I = Mk^2$	any axis
Rod, centre	$I = ML^2/12$	$\perp$ to rod
Rod, end	$I = ML^2/3$	$\perp$ to rod
Ring / bent rod	$I = MR^2$	central, $\perp$ plane
Ring, diameter	$I = MR^2/2$	in-plane
Sector of disc	$I = mr^2/2$	through disc centre, $\perp$ plane
Disc	$I = MR^2/2$	central, $\perp$ plane
Disc, diameter	$I = MR^2/4$	in-plane
Solid cylinder	$I = MR^2/2$	own axis
Hollow cylinder (thin)	$I = MR^2$	own axis
Hollow cylinder (thick)	$I = M(R_1^2 + R_2^2)/2$	own axis
Solid sphere	$I = 2MR^2/5$	diameter
Hollow sphere (thin)	$I = 2MR^2/3$	diameter
Rectangular plate	$I_C = M(a^2 + b^2)/12$	$\perp$ plane, centre
Triangular plate	$I_{base} = Mh^2/6$, $I_G = Mh^2/18$	in-plane, along/parallel base
Triangular plate, $\perp$	$I_{G,\perp} = M(a^2 + b^2 + c^2)/36$	through centroid
Perpendicular axis	$I_z = I_x + I_y$	plane lamina only
Parallel axis	$I_O = I_C + Mr^2$	one axis through CM
Torque	$\vec{\tau} = \vec{r} \times \vec{F}$, $\tau = Fd$	d = perp. dist. to line of F
Rotational 2nd law	$\tau_{net} = I\alpha$	fixed axis, or about CM
Couple	$\tau = Fd$	same about every point
Pure rolling	$v_{cm} = \omega R$, $a_{cm} = \alpha R$	no slipping
Rolling, top point	$v_{top} = 2v_{cm}$	ICR at contact
Contact pt. accel.	$a = \omega^2 R$, towards centre	pure rolling
KE of rolling body	$K = \frac{1}{2}Mv_{cm}^2(1 + k^2/R^2) = \frac{1}{2}I_{ICR}\omega^2$	pure rolling
Accel. on incline	$a = \frac{g \sin \theta}{1 + k^2/R^2}$	pure rolling, no slip
Friction on incline	$f_s = \frac{k^2}{k^2 + R^2} Mg \sin \theta$	pure rolling
Min. μ_s for rolling	$\mu_s \geq \frac{k^2}{k^2 + R^2} \tan \theta$	rolling begins to slip below this
Speed after height h	$v_{cm} = \sqrt{\frac{2gh}{1 + k^2/R^2}}$	energy method, no-slip
Toppling force	$F_{topple} = Mgb/(2h)$	about bottom outer edge
Slide vs topple	slides if $\mu_s < b/2h$; topples if $\mu_s > b/2h$	pushed block
Self-toppling	$\tan \theta_{topple} = b/(2H_{cm})$	tilting surface
Work by torque	$dW = \tau d\theta$, $P = \tau\omega$	fixed axis
Rotational KE thm.	$W_{net} = \Delta(\frac{1}{2}I\omega^2)$	fixed axis
Angular momentum	$L = I\omega$ (fixed axis); $\vec{L}_O = I_{cm}\vec{\omega} + M\vec{r}_{cm} \times \vec{v}_{cm}$	general plane motion
Torque- L relation	$\vec{\tau}_{net} = d\vec{L}/dt$	general, any fixed point/CM
Ang. mom. conservation	$I_1\omega_1 = I_2\omega_2$	$\tau_{net} = 0$

22 Keyword Decision Tree

- “ α constant” / find θ, ω at time $t \rightarrow$ rotational kinematics equations (Eq. 1.2)
- “uniform rod/disc/ring/sphere, about ...” $\rightarrow$ look up in the Master M.I. Table (Sec. 8.3); if the axis given is *not* the standard one, apply Parallel and/or Perpendicular Axis Theorem
- “plane lamina, about axis $\perp$ to the plane” and you know the two in-plane axes $\rightarrow$ Perpendicular Axis Theorem ($I_z = I_x + I_y$)
- “shifted axis, parallel to a centroidal axis” $\rightarrow$ Parallel Axis Theorem ($I_O = I_C + Mr^2$)
- “find the torque of a force” $\rightarrow \tau = Fd$ (perpendicular distance to line of action) or $\vec{\tau} = \vec{r} \times \vec{F}$
- “fixed axis, find angular acceleration” $\rightarrow \tau_{net} = I\alpha$ about that axis
- “rolling without slipping” $\rightarrow$ constraint $v_{cm} = \omega R$; for dynamics, pair $F_{net} = Ma_{cm}$ with $\tau_{cm} = I_{cm}\alpha$
- “acceleration of a rolling body down an incline” $\rightarrow a = \frac{g \sin \theta}{1 + k^2/R^2}$ (Eq. 5.7)
- “speed after rolling down height h ”, energy given $\rightarrow$ Eq. 5.8 (friction does no work in pure rolling)
- “minimum friction needed to roll” $\rightarrow$ Eq. 5.9
- “block pushed / on a ramp – does it slide or topple” $\rightarrow$ compare μ_s with $b/2h$ (Eq. 5.11) or $b/2H_{cm}$ (Eq. 5.12)
- “velocity of a point on a rolling wheel” $\rightarrow$ ICR at the contact point, $v = \omega \times$ (distance from ICR) (Eq. 5.4)
- “ I of a system changes, no external torque” $\rightarrow$ conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$
- “work done by a torque” / “power of a motor” $\rightarrow W = \tau\Delta\theta$, $P = \tau\omega$

23 Symbol & Sign Glossary

Key Points θ, ω, α = angular displacement, velocity, acceleration (rad, rad/s, rad/s²). I = moment of inertia (kg m²), always stated about a specific axis. k = radius of gyration, $I = Mk^2$. τ = torque (N m). L = angular momentum (kg m²/s). CCW = +ve, CW = -ve by the standard right-hand-rule convention used throughout this book.

v_{cm}, a_{cm} = velocity/acceleration of the centre of mass. ICR/IAR = instantaneous centre/axis of rotation — the point/line of zero velocity *at that instant only*; never differentiate $v = \omega r_{ICR}$ to obtain acceleration.

Pure rolling: static friction, magnitude found from equations (not necessarily $\mu_s N$); *does no work* (energy conserved). Rolling with slipping: kinetic friction = $\mu_k N$, direction opposes relative sliding, *does* (negative) work.

Last-minute exam traps

- Perpendicular Axis Theorem $\Rightarrow$ plane lamina **only**; Parallel Axis Theorem $\Rightarrow$ one of the two axes **must** pass through the centre of mass.
- $\tau = I\alpha$ about the CM is valid even while the CM accelerates; about any *other* moving point it is generally **not** valid unless that point is the ICR under special conditions – when in doubt, use the CM.
- Static friction in pure rolling is a constraint force: solve for it from the equations; do not assume it equals $\mu_s N$ unless the body is on the verge of slipping.
- The ICR’s velocity is zero, but its acceleration is generally **not** zero – do not use $a = \alpha r_{ICR}$.
- Toppling is decided by torque balance about the tipping edge (position of N), not by the size of F_{net} .
- Always keep θ in radians in every rotational formula; degree-to-radian conversion ($\times \pi/180$) is required whenever an angle is given in $^\circ$.

End of formula & derivation book — every boxed result above is traceable to a first-principles derivation from the definition of moment of inertia, Newton’s second law, or the definition of angular momentum.